
Task 3:-

DATA ANALYSIS PROJECT (PANDAS)

Goal:

Analyze a dataset, clean it, and generate summary statistics.

Requirements:

- Load CSV using Pandas
- Perform cleaning, filtering, grouping
- Generate meaningful insights

Deliverables:

1. Notebook (.ipynb) or script
2. Graphs (optional)

Project: Student Marks Data Analysis

Dataset (students.csv)

Create a CSV file named **students.csv** with the following data:

Name,Department,Marks

Alice,CSE,85

Bob,ECE,78

Charlie,CSE,92

David,EEE,65

Eva,CSE,88

Frank,ECE,74

Grace,EEE,90

Henry,CSE,81

Ivy,ECE,69

Jack,EEE,95

Python Script (data_analysis.py):

```
import pandas as pd
```

```
# Load CSV file
```

```
df = pd.read_csv("students.csv")
```

```
print("Original Dataset")
```

```
print(df)
```

```
# Data Cleaning
```

```
df = df.drop_duplicates()    # Remove duplicate rows
```

```
df = df.dropna()            # Remove missing values
```

```
# Filter students with marks >= 80
```

```
top_students = df[df["Marks"] >= 80]
```

```
print("\nStudents Scoring 80 or Above")
```

```
print(top_students)
```

```
# Group by Department
```

```
department_average = df.groupby("Department")["Marks"].mean()
```

```
print("\nAverage Marks by Department")
```

```
print(department_average)
```

```
# Summary Statistics
print("\nSummary Statistics")
print(df.describe())

# Highest Scorer
highest = df.loc[df["Marks"].idxmax()]
print("\nHighest Scorer")
print(highest)
```

Sample Output:

Original Dataset

	Name	Dept	Marks
0	Alice	CSE	85
1	Bob	ECE	78
2	Charlie	CSE	92
3	David	EEE	65
4	Eva	CSE	88
5	Frank	ECE	74
6	Grace	EEE	90
7	Henry	CSE	81
8	Ivy	ECE	69
9	Jack	EEE	95

Students Scoring 80 or Above

Alice

Charlie

Eva

Grace

Henry

Jack

Average Marks by Department

CSE 86.5

ECE 73.7

EEE 83.3

Summary Statistics

Count : 10

Mean : 81.7

Max : 95

Min : 65

Highest Scorer

Name: Jack

Department: EEE

Marks: 95

Insights:

- **Jack** scored the highest marks (**95**).
- The **CSE** department has the highest average marks (**86.5**).
- **6 students** scored **80 or above**.

- The overall average marks are approximately **81.7**.

Optional Graph (Bar Chart):

If you want to visualize the average marks by department, add the following code:

```
import matplotlib.pyplot as plt
department_average.plot(kind="bar")
plt.title("Average Marks by Department")
plt.xlabel("Department")
plt.ylabel("Average Marks")
plt.show()
```

Project Folder Structure:

```
Student_Data_Analysis/
|
|— students.csv
|— data_analysis.py
|— data_analysis.ipynb (optional)
|— graph.png          (optional screenshot)
└— README.md
```